INCH-POUND

MIL-A-52879C
21 April 1992
SUPERSEDING
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21 April 1980

MILITARY SPECIFICATION

ALARM, AUDIBLE, BZ-204/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an audible alarm, which is a local notification device for the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350

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PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-T-704 - Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-2000 - Standard Requirements for Soldered Electrical and Electronic Assemblies.
MIL-STD-45662 - Calibration System Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.

(Unless otherwise specified, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

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DRAWINGS

ME

TA13220E3700 - Alarm, Audible, BZ-204/FSS-9(V).
TA13220E3701 - Electronics Assembly.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JIE, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3953 - Strapping, Flat Steel and Seals (DoD adopted).
ASTM D 4675 - Selection and Use of Flat Strapping Materials (DoD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

D9.1 - Welding of Sheet Metal.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The audible alarm (hereinafter called "AA") shall be in accordance with Top Assembly TA13220E3700, and as specified herein.

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3.1.1 Major components. The AA shall contain the following major component:

a. Electronics Assembly (EA), TA13220E3701.

When specified (see 6.2), the EA shall be furnished separately.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the AA or EA when furnished separately. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The AA shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the AA may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the AA may be newly fabricated from recovered materials to the maximum extent practicable, provided the AA produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the AA.

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3.4 Environmental conditions. The AA and EA when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.2), performance degradation, or malfunction when tested as specified in 4.5.2.6.

3.4.1 Operational.

a. Temperature range of -20 °F to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Non-operational condition and storage.

a. Temperature range of -30 °F to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling; equipment not operating (without battery).

3.5 Detailed technical requirements. The AA and EA when furnished separately shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Tamper switch operation. The door tamper switch shall be mounted and adjusted such that its contacts open with not less than 0.125 inch or more than 0.250 inch of door movement and before the door flanges are disengaged. The pry-off tamper switch shall be so mounted that with the AA on its back on a flat surface its contacts open when the hinged side of the AA is raised not less than 0.125 nor more than 0.250 inch. For both switches, the contacts shall also be closed when the shaft is pulled all the way out to a detented position hereinafter referred to as the maintenance position.

3.5.2 Terminal board connections. All terminal boards and terminals shall be labeled. The function of each terminal on terminal boards that interface with external devices shall be as follows:

TB1	Function
1	Power, ac (phase)
2	Power, ac (neutral)
3	Ground

TB2	Function
1	Alarm input
2	Alarm and tamper common
3	Tamper output
4	Spare
5	Spare

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3.5.3 Primary power supply. The primary power supply shall operate from 110-125 volts, 48-62 hertz (Hz) alternating current (ac) power. The primary power supply shall provide nominal 27.6 volts direct current (Vdc) power for float charging of a 24 Vdc nominal, 2.5 ampere hour battery and for circuit operation.

3.5.3.1 Voltage output. The voltage provided at the battery leads with the battery disconnected shall be 27.6 ±0.5 Vdc at an ambient temperature of 76 ±8 °F, 29.9 ±1.2 Vdc at -20 °F, 25.6 ±1.2 Vdc at 150 °F, and 27.6 ±1.0 during humidity and salt fog tests. With a 260 ohm resistor connected in place of the battery across the battery leads, the voltage shall be 24.8 ±0.5 Vdc at 76 ±8 °F.

3.5.3.2 Fast charge circuitry characteristics. The power supply shall provide fast charge circuitry which acts to provide a greater charging current when the voltage across the battery terminals falls below 24.0 ±0.5 Vdc (24.0 ±1.0 Vdc at -20 °F and +150 °F). With a 100 ohm resistor connected in place of the battery across the battery leads, the voltage shall vary between 20 ±3 volts and 32 ±3 volts with a period of 8.3 ±1 milliseconds (ms).

3.5.4 Emergency power supply. The emergency power supply shall be a 24 volt, sealed, gelled electrolyte, rechargeable battery floating on the output of the battery-charging network. The battery is not a deliverable item. The current drain presented by the AA to an emergency power supply providing 24 Vdc shall be 26 ±6 milliamperes (ma) with no tamper conditions present and a no alarm input (see 3.5.5.1) to the AA and 1.0 ±0.15 amperes when an alarm input condition (see 3.5.5.1) is present.

3.5.5 Alarm and tamper alarm characteristics.

3.5.5.1 Alarm operation. The AA shall be in the no alarm state when a voltage of 4 Vdc and greater is applied between the alarm input and common terminals (see 3.5.2). The AA shall be in the alarm state when the applied voltage across these terminals is less than 1.0 Vdc.

3.5.5.2 Alarm output using dummy load. The output of the AA shall have the following characteristics:

a. With the speaker leads disconnected and an 8 ohm resistive load substituted across TB3-6 and -7, the voltage output shall be 13.5 ±1.0 volts true root mean square (RMS) when the AA is in the alarm state.
b. The output waveform for the above condition shall be a varying frequency distorted square wave, sweeping from 850 ±30 percent to 2300 Hz ±30 percent with a period of 0.60 ±0.20 seconds. Amplitude spectrum shall be within 3 dB over the frequency range.

3.5.5.3 Tamper alarm operation. The AA shall signal a non-tamper alarm output by providing a voltage of 5.0 ±0.5 Vdc into a load of 3.0 kilohms at terminal TB2-3 referenced to TB2-2 and shall signal a tamper alarm output by providing less than 1 Vdc at these terminals. The tamper alarm shall be

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generated when either tamper switch is open, when the speaker is sounding, or when the resistance between the AA housing and AA enclosure (i.e. inner and outer housings) is less than 7.5k ohms.

3.5.6 Sound output. The AA, with door closed, shall comply with the following requirements when operating on either primary or emergency power:

a. The alarm sound level shall be a minimum of 110 decibels (measured with weighing Scale A, based on a reference level of 0.0002 microbars) (dB(A)) at a distance of 10 feet on the acoustic axis.
b. The characteristic sound of the alarm shall be a nonsinusoidal modulated tone swept through a frequency range of approximately 850 to 2300 Hz ±30 percent with a period of 0.55 ±0.2 seconds.

3.5.7 Electronics assembly. When the EA, when furnished separately, is installed in an AA whose other components are properly functioning, the AA shall meet the requirements specified in 3.5.3 through 3.5.6.

3.6 Burn-in. Each AA and EA when furnished separately shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Reliability. The AA shall demonstrate a mean-time-between-failure (MTBF) of 10,000 hours or greater when tested as specified in 4.5.2.9.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Identification marking. The AA and EA shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the AA normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G as applicable. Color shall be lusterless gray conforming to FED-STD-595, color No. 36440. Nonpermissible galvanic couples (dissimilar metals) as specified in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.6) for manufacturing and testing the AA and EA when furnished separately:

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Item No.	Description	Identification	Quantity
1	Alarm, Audible	BZ-204/FSS-9(V), NSN 6350-00-228-2514	5 (see note)
2	Stable base drawing of PWB Subassembly, Amplifier	13220E3744, Sh 2	1
3	Mylar drawing of PWB, Power Supply	13220E3763, Sh 2 & 3	1

Note: Provided only when EA's are procured separately.

3.12 Workmanship. Workmanship shall be in accordance with the requirements and guidance paragraphs of MIL-STD-454, requirement 9.

3.12.1 Soldering. All soldering shall be performed in accordance with the general requirements of MIL-STD-2000.

3.12.2 Welding. Sheet metal welding shall be in accordance with AWS D9.1, except no visible pores shall be allowed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor shall be responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.2 Classification of inspections. The AA and major components when furnished separately shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article AA and EA when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Complete AA's. Complete AA's shall be subjected to the tests marked "X" in column 1 of table I, except as noted. Failure of any test shall be cause for rejection.

4.3.2.2 EA's when furnished separately. EA's when furnished separately shall be subjected to the tests marked "X" in column 1 of table II, except as noted. Failure of any test shall be cause for rejection.

4.3.2.3 Measurement recording requirements. All measurements, including voltages, currents, resistances, times, and distances, observed during first article tests shall be recorded on the test data sheets and shall be to the precision necessary to demonstrate compliance with requirements in the appendix test method being conducted.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for tests shall be in accordance with MIL-STD-105 and as specified herein.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. Sample size shall be determined by using MIL-STD-105, table I and table II-a. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 AA's. Each AA shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

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4.4.3.1.2 EA's when furnished separately. Each EA when furnished separately shall be subjected to the tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 AA's. For normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table I. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced inspection shall be three units. For each sample size, the lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3.2.2 EA's when furnished separately. For normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table II. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection, and production is at a steady rate. Sample size during reduced inspection shall be three units. For each sample size, the lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The AA and the EA when furnished separately shall be examined as specified herein for the following defects (as applicable):

101. Electromagnetic interference suppression components and networks not as specified (see 3.1, 3.1.1, and 3.1.2).
102. Components missing or not as specified (see 3.1, 3.1.1, and 3.1.2).
103. Assembly not correct (see 3.1, 3.1.1, and 3.1.2).
104. Materials not as specified (see 3.3).
105. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
107. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
108. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the AA or EA when furnished separately (see 3.3.2).
109. Interchangeability not as specified (see 3.8).
110. Identification or special marking missing, incomplete or illegible (see 3.9).
111. Treatment and painting not as specified (see 3.10).

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112. Workmanship not as specified (see 3.12).
113. Soldering not as specified (see 3.12.1).
114. Welding not in accordance with AWS D9.1 (see 3.12.2).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient humidity conditions at a room ambient temperature of 76 ±8 °F. After successful completion of a functional test (see 4.5.2.5), each AA and EA when furnished separately shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the AA and EA shall operate as specified herein without maintenance.

4.5.2.2 Test schedule.

Table I. Test schedule for complete AA .

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	3.5.3, 3.5.4 3.5.5	4.5.2.5
X	X	-	Tamper switch tolerance	3.5.1	4.5.2.6
X	-	X	Sounding Device	3.5.6	4.5.2.7
X 1/	-	X	Low Temperature	3.4.1(a), 3.4.2(a)	4.5.2.8.1
X 1/	-	X	High Temperature	3.4.1(a), 3.4.2(a)	4.5.2.8.2
X 1/	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.5.2.8.3
X 1/	-	-	Shock	3.4.2(c)	4.5.2.8.4
X 2/	-	-	Reliability	3.7	4.5.2.9

1/ Reliability test units not subjected to this test.
2/ Environmental test units not subjected to this test.

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Table II. Test schedule for EA (when furnished separately). 1/

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	3.5.3, 3.5.4, 3.5.5	4.5.2.5
X	-	X	Low temp	3.4.1(a) 3.4.2(a)	4.5.2.8.1
X	-	X	High temp	3.4.1(a), 3.4.2(a)	4.5.2.8.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.5.2.8.3
X	-	X	Shock	3.4.2(c)	4.5.2.8.4

1/ The EA shall be placed in a government-loaned AA and tested in accordance with the applicable AA test method.

4.5.2.3 Test methods. Test shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each AA and EA when furnished separately shall be subjected to a 24-hour burn-in after power supply alignment (see appendix alignment procedure) and successful completion of the functional test (see 4.5.2.5). During burn-in, the AA shall be energized at the primary power specified. Eight times during the burn-in period, but not more often than once per hour, circuits shall be exercised by performing steps 3.a through 3.j of the reliability test (appendix test method 4), except as follows:

a. If desired, the speaker leads into TB3-6 and TB3-7 may be disconnected and an 8 ohm dummy load connected across these terminals. Presence of a low level sound (caused by transformer windings) should be observed in step 3.b.

b. The oscilloscope called for in steps 3.g through 3.i may be omitted; it will be sufficient to observe that the voltage fluctuates with switch S3 closed. A complete functional test shall be performed after completion of burn-in. Non-compliance with the applicable portions of 3.5 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be

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forwarded to the contracting officer or as designated. Any unit failing during burn-in or the post burn-in functional test shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the functional test.

4.5.2.5 Functional test. The AA and EA when furnished separately shall be tested in accordance with appendix test method 1. Nonconformance to 3.5.3, 3.5.4, and 3.5.5 shall constitute failure of the test.

4.5.2.6 Tamper switch tolerance. The AA door and pry-off tamper switches shall be tested in accordance with appendix test method 3. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.7 Sounding device. The AA shall be tested in accordance with appendix test method 2. Nonconformance to 3.5.6 shall constitute failure of this test.

4.5.2.8 Environmental. The AA and EA when furnished separately shall be subjected to the following environmental tests. General testing requirements shall be in accordance with the test methods section of MIL-STD-810, except standard ambient temperature shall be 76 ±8°F and the specified environmental test report is not required. The operating test required as part of the "pretest performance record", the operational checkout after installation in the chamber, the "performance check during test" at the operating temperature or humidity condition, and the operational checkout required as part of "post-test data" shall be the functional test specified in 4.5.2.5. During storage and operation, the AA shall be configured as deployed, except, during high and low temperature tests, the AA door may be left open to the minimum extent necessary to bring out wiring required to perform the operational checks. The AA shall not be removed from the environmental chamber for test at operating temperature for the high and low temperature tests.

4.5.2.8.1 Low temperature. Low temperature test shall be conducted in accordance with method 502.3. After performing the "pretest standard ambient checkout", perform procedure I, using storage temperature of -30 °F for not less than 48 hours. Perform procedure II using equipment operating temperature of -20 °F. Procedure III is not required.

4.5.2.8.2 High temperature. High temperature test shall be conducted as specified herein. Apparatus shall be in accordance with "Apparatus" section of method 501.3 of MIL-STD-810 except as specified herein. Test procedure shall be as follows:

a. Perform "pretest standard ambient checkout" in accordance with "Preparation for test" section of method 501.3, except do not proceed to "Procedures" section.
b. After completion of operational checkout of step a. above, raise internal chamber temperature to 120 °F.
c. Maintain internal chamber temperature at 120 °F for 6 hours.

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d. Raise internal chamber temperature to 165 °F within 1 hour and maintain temperature at 165 °F until 5 hours have elapsed since completion of step c.
e. Lower internal chamber temperature to 120 °F within 1 hour. Do not proceed until 1 hour has elapsed.
f. Repeat steps c, d, and e two more times for a total of three 12-hour cycles.
g. Raise chamber temperature to operating temperature of 150 °F and maintain for 1 hour or until temperature is stabilized, whichever is longer.
h. Operate the AA and conduct functional test specified in 4.5.2.5.
i. Return the AA, nonoperating, to standard ambient conditions and stabilize.
j. Operate the AA and conduct functional test specified in 4.5.2.5.

4.5.2.8.3 Humidity. Humidity test shall be conducted in accordance with method 507.3 except as specified. During storage and operation, the AA shall be configured as deployed. The AA door shall be closed during humidity cycling, except as specified herein. Apparatus shall be in accordance with "Apparatus" section of method 501.3. Test procedure shall be as follows:

a. Perform "pretest standard ambient checkout" in accordance with "Preparation for test" section of method 507.3. Prior to starting the test, the chamber shall be at 76 ±8°F with uncontrolled humidity.
b. Gradually raise the internal chamber temperature to 149 °F and the relative humidity to 95 percent or greater over a period of 2 hours.
c. Maintain conditions of step b for 6 hours.
d. Gradually reduce internal chamber temperature in 16 hours to 86 °F, always maintaining 92 percent or greater relative humidity.
e. Repeat steps b, c and d for a total of 5 cycles (120 hours minimum). Perform the functional test before the completion of each cycle.
f. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. The doors of the AA shall not be opened during measurements. The AA may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the AA to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.5.2.8.4 Shock. Shock test (bench handling) shall be performed in accordance with method 516.4, procedure VI.

4.5.2.8.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.9 Reliability. Using an upper test MTBF 0_0 of 10,000 hours and a lower test MTBF 0_1 of 5,000 hours, the AA shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-HDBK-781, test plan VIIID. Continue testing until either an "accept" or "reject" decision is

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reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage (see 6.4.2) to the system.
c. Serious personnel safety hazards.

4.5.2.9.1 Test procedure. One or more AA's shall be randomly selected by a Government representative from AA's that have successfully completed the 24 hour burn-in and functional test. The testing shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Equipment shall be energized at the primary power input specified (see 3.5.3). Operation shall be continuous throughout the test with circuits exercised not less than nine times during each 24-hour period, but not more often than once per hour. The test shall be conducted in accordance with appendix test method 4. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine (false alarms) (see 6.4.1) shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. No adjustments shall be allowed during the test.

4.5.2.9.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table II-a. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

115. AA's not preserved and keys not placed with the AA as specified for levels A, B, or C (see 5.1).
116. EA's furnished separately not preserved and consolidated as specified for levels A, B, or C (see 5.1).
117. Complete AA's and EA's not packed in a shipping container as specified for levels A or B (see 5.2).
118. Cushioning, blocking and securing within the individual box not as specified for levels A and B (see 5.2).
119. Strapping not as specified for levels A and B (see 5.2).

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120. Quantities packed together exceed the size or weight limitations of the box for levels A and B (see 5.2).
121. Marking missing, illegible, incorrect, or incomplete for levels A, B, or C (see 5.3).

5. Packaging.

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each AA and EA when furnished separately shall be preserved by one of the following procedures depending on procurement of complete AA's or EA's furnished separately.

5.1.1.1 Procurement of complete AA's. Each AA shall be preserved in accordance with MIL-P-116, submethod IIb. The inner container shall conform to PPP-B-636, type CF, class Domestic, variety SW or DW, grade 200. The AA shall be cushioned and secured within the container to prevent movement or damage. The keys shall be placed in a cotton draw-string bag or heat-sealed bag and placed within the container. The outer container shall conform to PPP-B-636, W5c, style optional.

5.1.1.2 Procurement of EA's when furnished separately. Each EA when furnished separately shall be preserved in accordance with MIL-P-116, submethod IIb. The inner container shall conform to PPP-B-636, type CF, class Domestic, variety SW, grade 175. The EA shall be cushioned and secured within the container to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c or W5c, style optional.

5.1.1.2.1 Consolidation. EA's, preserved as specified in 5.1.1.2, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional, in quantities not to exceed the weight limitations of the box. The contents shall be cushioned within the box to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except submethod IIc shall be used in place of submethod IIb. Each AA or EA when furnished separately shall be placed in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional. The components shall be cushioned within the box to prevent movement or damage.

5.1.3 Level C. Each AA with its keys shall be preserved together. EA's when furnished separately shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first using activity. As a minimum, each AA or EA when furnished separately shall be placed in a fiberboard box.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The AA and EA's when furnished separately, preserved as specified in 5.1, shall be packed by one of the following procedures depending upon procurement of complete AA's or procurement of EA's when furnished separately.

5.2.1.1 Procurement of complete AA's. AA's preserved as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the size and weight limitations of the box. Cushioning and blocking within the box used to prevent movement and isolate impact shock shall be in accordance with MIL-STD-1186. Closure shall be in accordance with the appendix to the box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.1.2 Procurement of EA's when furnished separately. EA's, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete AA's and EA's when furnished separately, preserved as specified in 5.1, shall be packed as specified in 5.2.1 as applicable, except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be as specified for level A except it may be zinc-coated.

5.2.3 Level C. Each AA and EA's when furnished separately, preserved as specified in 5.1, shall be packed in shipping containers that will afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.2.

5.3.2 Special marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AA is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Nomenclature and quantity of item(s) to be furnished (see 3.1.1).
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval, the number of units required, and the allocation of required units to environmental and reliability tests respectively (see 3.2).
f. Treatment and painting if other than specified (see 3.10).
g. When Government-loaned property is not required (see 3.11).
h. Schedule, content, and format of burn-in test reports (see 4.5.2.1).
i. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of five units for environmental tests and twelve units for reliability tests. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

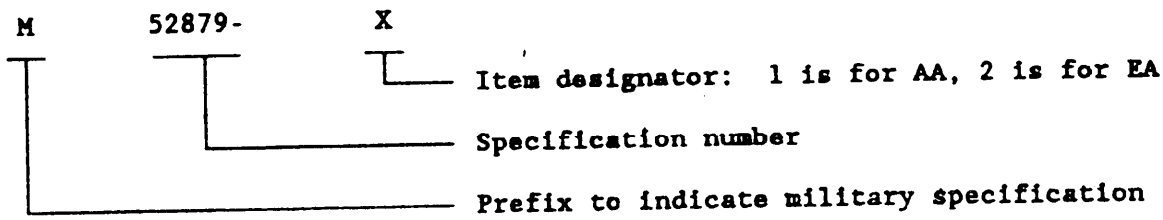
6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the AA; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.6 Government-loaned property. The contracting officer should arrange to lend the property specified in 3.11 when applicable (see 6.2).

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6.7 Part or identifying number (PIN). The PINs to be used for the AA or EA, acquired to this specification, are created as follows:



Example: M52879-1 is PIN for AA.

6.8 Subject term (key word) listings.

Audible signaling device
Intrusion detection
Physical security

6.9 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - EC
Air Force - 99

Review activities:
Navy - MC
DLA-GS

Preparing activity:
Army - ME

Project 6350-0126

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APPENDIX

TEST METHODS

FOR

AA

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether AA's procured under this specification conform to the requirements specified. Figures A-1 through A-4 are required to facilitate testing of the AA. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

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APPENDIX

TEST METHOD 1

FUNCTIONAL TEST

1. TEST APPARATUS:

a. Power Supply (PS1) - Constant voltage, continuously adjustable, 0-6 Vdc, 3 ma minimum, line regulation < 1 percent, load regulation (no load to full load) < 1 percent, peak to peak ripple voltage < 50 millivolts (mV).
b. Power Supply (PS2) - Constant voltage, continuously adjustable, 0-30 Vdc, 2 A minimum, line regulation < 0.05 percent, load regulation (no load to full load) < 0.1 percent + 3 mV, peak to peak ripple voltage < 5 mV.
c. Voltmeter, true RMS (M1) - minimum resolution 0.2 V RMS at voltages up to 15 V RMS, minimum accuracy 2 percent, frequency range from < 500 to > 3000 Hz.
d. Voltmeter, digital, dc (M2, M3, M4) - 3-1/2 digit or greater, minimum accuracy 0.5 percent of reading ±1 least significant digit (LSD).
e. Voltmeter, digital, dc (M5) - 3-1/2 digit or greater, minimum accuracy 0.2 percent of reading ±1 LSD.
f. Ammeter (M6) - 0-2 Adc, minimum resolution 0.1 A, minimum accuracy 2 percent, maximum voltage drop (burden voltage) 0.3 Vdc.
g. Oscilloscope - General purpose, as required for resolution and accuracies specified in this procedure.
h. Spectrum analyzer - Bruel & Kjaer 4033 or equivalent.
i. Dummy load (R1) - 8 ±0.2 ohms, 25 watts minimum.
j. Resistor (R2) - 3K ohm ±1 percent, 1/4 watt.
k. Resistor (R3) - 3K ohm ±20 percent, 1/2 watt.
l. Resistor (R4) - 7.5K ohm ±1 percent, 1/4 watt.
m. Resistor (R5) - 160 ohm ±1 percent, 1/4 watt.
n. Resistor (R6) - 100 ohm ±1 percent, 1/4 watt.
o. Diode (CR1) - 1N4002 or equivalent.
p. Transistor (Q1) - MK2955 or equivalent.
q. Switch (S1) - 1 pole, 3 position, 75 V, 1.5 A minimum.
r. Switch, (S2) - SPDT, 75 V, 1 A minimum.
s. Switch (2 required) (S3, S4) - SPST, 50 V, 0.5 A minimum.
t. Power cord and plug, 115 Vac.
u. Test set-up (see figure 1).

NOTE: Number of required meters may be reduced by use of suitable switching arrangements or by moving test leads as necessary.

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APPENDIX

2. TEST PREPARATION:

a. Connect test equipment as shown in figure 1. Note that the leads from the speaker to TB3-6 and TB3-7 are to be disconnected and an 8 ohm dummy load (R1) connected across TB3-6 and TB3-7. The resistance of the leads from TB3-6 and -7 to the dummy load shall be no more than 0.2 ohm. The wires connected to the inner and outer housings may be attached to any stable points that make good electrical contact with the housings.
b. Set S1 to position 1 and S2, S3, and S4 to open.
c. Turn on PS1 and adjust it to read 4.0 ±0.2 Vdc on M2.
d. Turn AA power switch to the ON position. Do not apply 117 Vac to AA power plug until instructed to do so.
e. Place the pry-off and door tamper switches in fully depressed position or maintenance position (all the way out).

3. TEST PROCEDURE:

NOTE:

All required measurements are to be recorded during all first article tests. Check marks may be used to denote compliance with specified parameters during production tests .

a. Apply 117 ±7 Vac to AA power plug. Observe and record M1 and M3 readings. M1 shall indicate 0 Vac and M3 shall indicate 5.0 ±0.5 Vdc.
b. Reduce PS1 voltage to 0.9 Vdc. M1 shall indicate 13.5 ±1.0 volts rms and M3 shall indicate less than 1 Vdc. Record values.
c. Observe pattern on spectrum analyzer. The pattern shall show frequency content over the range from 850 Hz (±30 percent) to 2300 Hz (±30 percent) with amplitude variation over the range not exceeding 3 dB. Record frequencies at the 3 dB points.
d. Using time function on spectrum analyzer or using an oscilloscope, observe period of sweep. It shall be 0.6 ±0.2 second. Record period.
e. Adjust PS1 to 4.5 ±0.5 Vdc. M1 and M3 shall indicate as in step a above.
f. Close switch S4 to connect R4 from the inner housing to the outer housing.
g. Measure the voltage on M3. It shall be less than 1.0 Vdc.
h. Open switch S4. M3 shall indicate as in step a above.
i. If the AA has been operating for at least 10 minutes, proceed to step j. Otherwise allow additional time as necessary before proceeding.

NOTE: AA may be warmed up immediately prior to testing to avoid delay at this point in testing.

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j. Observe the voltage on M5. It shall be 27.6 ±0.5 Vdc during room temperature (76° ±8 °F) testing, 29.2 ±1.2 Vdc at -20 °F and 26.0 ±1.2 Vdc at +150 °F.

k. Set oscilloscope for ac coupled input at a sensitivity of 0.2 volts/cm and a sweep rate of 0.002 sec/cm or as desired.

l. Turn on power supply PS2 and adjust for a voltage on M4 equal to or slightly greater than the voltage just recorded in step j above. If the voltage of step j exceeded 30 Vdc, set PS2 for 30 Vdc.

m. Close switch S2.

n. Slowly decrease voltage of PS2 while carefully monitoring M5 and oscilloscope. Observe the voltage on M5 just before the scope trace suddenly shows a distorted full wave rectified voltage 0.1 volt peak or greater. It shall be 24.0 ±0.5 Vdc at 76 ±8 °F, 24.0 ±1.0 Vdc during environmental tests.

o. Increase PS2 voltage until ac voltage seen on scope in step n above ceases and continue to increase PS2 until M5 indicates a voltage several tenths of a volt greater than that of step n above when the rectified voltage began.

p. Decrease PS2 voltage very slowly until the waveform of step n re-appears. Observe the voltage on M5 at the transition point. It shall be 24.0 ±0.3 Vdc at 76 ±8 °F, 24.0 ±1.0 Vdc at all other temperatures. If voltage differs from that of step n by more than 0.1 Vdc, repeat steps o and p until consecutive readings within 0.1 Vdc are obtained. Record this value.

q. Open switch S2 and then set S1 to position 2.

r. Set oscilloscope for dc coupled input and set vertical sensitivity to allow viewing of dc input levels up to 30 Vdc. Scope shall show a steady dc voltage and meter M5 shall read 24.8 ±0.5 Vdc. Record M5 reading.

s. Close S3. Oscilloscope shall show a voltage varying between 20 ±3 volts and 32 ±3 volts with period of 8.3 ±0.5 milliseconds. Record voltage maximum and minimum readings. Scope settings may be adjusted as necessary to facilitate reading.

t. Adjust PS2 for output voltage of 25 Vdc on M4.

u. Set S1 to position 3.

v. Remove 117 Vac power into AA. There shall be no change in the readings on M1 and M3.

w. Adjust PS2 as necessary to read 24 Vdc on M5.

x. Observe and record current on M6. It shall be 26 ±8 mA.

y. Turn off PS1. Adjust PS2 as necessary to maintain 24 ±0.3 Volts dc (average) on M5. Observe current on M6 and voltage on M1. M6 shall indicate 1.0 ±0.15 amperes and M1 shall indicate 11.0 ±0.5 volts RMS. Record values.

z. Adjust PS2 to read 20 ±0.5 Volts dc (average) on M5. Current on M6 shall be 0.85 ±0.10 amperes and voltage on M1 shall be 9.0 ±0.5 volts RMS. Record values.

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aa. Turn off all power supplies, remove dummy load and reconnect speaker, remove R4 and other connections used for testing, except do not remove source of ac power. Insulate terminals of battery leads as necessary to assure they do not make contact with wires or enclosures.

bb. Using protective earplugs or suitable sound deadening cover over the AA verify that the AA sounds by momentarily turning on AC power.

cc. Turn off 117 VAC source and remove ac connection to AA. Remove any insulating material applied to battery lead terminals.

NOTE: Steps aa, bb and cc need not be performed if a subsequent functional test is to be performed on the same AA.

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APPENDIX

TEST METHOD 2

SOUND LEVEL OUTPUT TEST

1. TEST APPARATUS:

   a. DC power source - +5 ±1.0 Vdc, 2 mA minimum.
   b. AC power cord and plug.
   c. Sound pressure level meter, consisting of following (or equivalent system):
      (1) Octave band analyzer - General Radio (GR) Type 1558 or equivalent.
      (2) Pre-amplifier and microphone set - GR model 1560-P40K or equivalent.
      (3) Cable assembly (25 feet) - GR model 1560-P72E or equivalent.
   d. Battery - Drawing 13220E3712 (included within TA 13220E3700).
   e. Test chamber (see figure 2).

2. TEST PREPARATION:

   NOTE: The sound level output test shall be performed using a sound chamber equivalent to that described in figure 2.

   a. Install the AA under test inside the test chamber.
   b. Connect wires from the 5 Vdc power source to terminals TB2-1 (+) and TB2-2 (-). Wires should be brought through a mounting hole in back of AA.
   c. Connect an ac power cord through a mounting hole in back of AA to TB1-1, TB1-2 and TB1-3 (ground). Do not apply power to ac cord.
   d. Install battery within AA.
   e. Set up sound level meter on a common axis to that of the speaker 10 feet in front of unit. Set the band level dial at the 110 dB position, the band switch to "A" (weighing scale A), and the 6 position switch to slow (slow response meter setting).
   f. Turn on the 5 Vdc power source.
   g. Apply 117 Vac power to the ac power cord and then turn AA power switch to on position.
   h. Secure the inner housing cover and close and lock the outer housing door.

3. TEST PROCEDURE:

   WARNING: A suitable sound barrier or earplugs must be worn at all times while testing the unit. The sound level this unit is capable of producing can cause permanent hearing damage.

   a. Turn off power from 5 Vdc power source. AA shall sound.

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b. Observe and record the sound pressure level (minimum reading). Sound pressure level shall be 110 dB or greater.

NOTE: If level is not 110 dB or greater, turn on 5 Vdc power and check to see that battery is fully charged. When AA is not sounding and battery is on float charge within the AA, a good, well charged battery will draw less than 20 mA and the voltage across its terminals should be at least 26.5 Vdc.

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APPENDIX

TEST METHOD 3

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

a. Power source (PS1) - +5 ±1.0 Vdc, 3 mA minimum.
b. Voltmeter, dc (M1) - resolution 0.2 Vdc at 5 Vdc, accuracy 3 percent of reading.
c. Plate gage, 0.125 inch thick.
d. Plate gage, 0.250 inch thick.
e. Ac cord and plug (optional)
f. Battery (optional) - Drawing 13220E3712 (included within TA 13220E3700).
g. Resistor (R1) - 3K ohm ±1 percent, 1/4 watt.
h. Tamper switch tolerance test setup (see figure 3).

NOTE: Either e. or f. must be provided.

2. TEST PREPARATION:

a. Connect test equipment to AA as shown in figure 3. Either connect ac power cord to TB1-1 (phase), TB1-2 (neutral) and TB1-3 (ground) or connect battery to battery leads. If battery is not used, insulate both leads to prevent their touching either housing.

NOTE: Shorting the red lead to either housing may cause component failure; Shorting the black lead to the inner housing will cause a tamper alarm.

b. Before turning AA power switch on be sure 5 Vdc is being provided from TB2-1 to TB2-2 to prevent the AA from sounding during the test.
c. Place the AA power switch in the ON position.
d. Secure the inner housing cover and close and secure the outer door.
e. If AC power is being used, connect plug to source.

3. TEST PROCEDURE:

a. Place the unit on a flat surface in a manner to hold the pry-off tamper switch closed.
b. Measure the voltage reading from TB3-3 to TB3-2. This voltage shall be 5 ±0.5 Vdc.
c. Slide a 0.125 inch thick plate gage between the midpoint of the door hinge edge and the flat surface. Measure the voltage. This voltage shall be 5 ±0.5 Vdc.

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d. Repeat step c at the same location with a 0.250 inch thick plate gage. Measure the voltage. This voltage shall be less than 1.0 Vdc.
e. Remove the 0.250 inch thick gage.
f. Measure the distance (x) from the flat surface to the door flange.
g. Increase the distance to (x +0.125) inch by opening the door 0.125 inch. Measure the voltage. It shall be 5.0 +0.5 Vdc.
h. Increase the distance to (x +0.250) inch by opening the door 0.250 inch from the closed position. Measure the voltage. It shall be less than 1.0 Vdc.
i. Remove unit from flat surface sufficiently to place pry-off tamper switch in maintenance position. Open door and place door tamper switch in maintenance position. Measure the voltage. Voltage shall be 5 $\pm$5 Vdc.

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APPENDIX

TEST METHOD 4

RELIABILITY TEST

1. TEST APPARATUS:

a. DC power source - 4.5 ±0.5 Vdc, 2 mA minimum.
b. AC power cord and plug.
c. Voltmeter, digital, dc (3 required) (M1, M2, M3) - 3-1/2 digit or greater, minimum accuracy 0.5 percent of reading ±1 LSD.
d. Oscilloscope - General purpose, as required for resolution and accuracies specified in this procedure.
e. Resistor (R1) - 7.5K ohm ±1 percent, 1/4 watt.
f. Resistor (R2) - 3K ohm ±1 percent, 1/4 watt.
g. Resistor (R3) - 160 ohm ±1 percent, 1/4 watt.
h. Resistor (R4) - 100 ohm ±1 percent, 1/4 watt.
i. Switch (2 ea) (S3,S4) - SPST, 50 V, 0.5 A minimum.
j. Reliability Test set-up (see figure 4).
k. Functional Test set-up (see figure 1 (see note)).

NOTE: The test set-up of figure 1 and equipment of appendix test method 1 are required once daily.

2. TEST PREPARATION:

CAUTION: The AA produces an output sound pressure level in excess of 110 dB which is sufficient to cause permanent hearing loss. Personnel in the test area must wear suitable ear protection.

a. Connect test equipment as shown in figure 4.
b. Adjust DC power source for output of 4.5 ±0.5 Vdc.
c. Close S2 and verify M1 indicates 4.5 ±0.5 Vdc.
d. Place the pry-off tamper switch in fully depressed position.
e. Set S1 and S3 to open.
f. Turn AA power switch to the ON position and close and secure AA door.
g. Apply 117 Vac (nominal) to AA by plugging in AC power plug to source of 117 Vac.

3. TEST PROCEDURE:

NOTE: The reliability test begins at step 2.g of test preparation. During the reliability test, output TB2-3 shall be continuously monitored. Any alarms that occur other than those specifically required by the test procedure shall be noted and explained in the test report. Eight times during each 24 hours, but not more than once per hour, steps a through j below shall be performed. At the end of each 24 hour period, the complete functional test

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APPENDIX

in accordance with appendix test method 1 shall be conducted and, in addition, steps k through n below shall be performed. AA's may be removed from the test fixture to perform the once daily functional test. Except where specified during these test periods, 117 Vac (nominal) power shall be continuously applied to the AA, the AA power switch shall be on, and the door shall be closed. Test periods are considered as part of the total elapsed time of the reliability test, provided that no more than 2 minutes per unit per day are expended with both ac and dc power to the AA removed or with the AA power switch off. If additional time is required without power the length of test must be extended accordingly.

a. Measure the voltage on M2. M2 shall indicate 5.0 ±0.5 Vdc.
b. Open S2 to for at least 5 seconds. AA shall sound and M2 shall indicate less than 1 Vdc while S2 is open.

NOTE: It is not necessary to measure for sound level output; however any abnormality noted in the level or frequency of the sound shall require that the appropriate test be performed (appendix test method 2 if power level is low, steps 3.b, 3.c, 3.d, and 3.e of appendix test method 1 if frequency or repetition rate is abnormal) to verify the audible output is within specification.

c. Close S2. AA shall be silenced and M2 shall indicate 5.0 ±0.5 Vdc.
d. Close S1 to connect R4 (7.5K ohm resistor) from the inner housing to the outer housing. AA shall be silent.
e. Measure the voltage on M2. It shall be less than 1.0 Vdc.
f. Open S1. M2 shall indicate 5.0 ±0.5 Vdc.
g. Place scope leads across red and black battery leads (probe to red).
h. Set oscilloscope for dc coupled input and set vertical sensitivity to allow viewing of dc input levels up to 35 Vdc. Scope shall show a steady dc voltage and meter M3 shall read 27.0 ±0.5 Vdc.
i. Close S3. Oscilloscope shall show a voltage varying between 20 ±2 volts and 32 ±2 volts with period of 8.3 ±0.5 milliseconds. Scope settings may be adjusted as necessary to facilitate reading.
j. Open S3. M3 shall indicate 27.0 ±0.5 Vdc.

NOTE: Step j above completes the test required eight times per day. The following steps k through n are to be performed daily in addition to the functional test of appendix test method 1.

k. Open AA door. AA shall not sound and M2 shall indicate less than 1 Vdc.
l. Close AA door. M2 shall indicate 5.0 ±0.5 Vdc.
m. Raise AA as required to activate pry-off tamper switch. AA shall not sound and M2 shall indicate less than 1 Vdc.
n. Return AA to its normal position. M2 shall indicate 5.0 ±0.5 Vdc.

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APPENDIX

AUDIBLE ALARM ALIGNMENT PROCEDURE

NOTE: This procedure is performed with the cover of the Electronic assembly removed to provide access to the power supply pwb assembly. This procedure shall be performed at a room temperature of 76 ±8 °F.

a. Set up in accordance with figure 1 as modified in this paragraph.
b. Set pot R11 on power supply pwb fully clockwise.
c. Assure AA switches S1 (door tamper) and S2 (pry off tamper) are closed.
d. Temporarily install resistor R1 (any value from table on pwb assembly drawing) or use decade resistor box.
e. Insulate battery connection wires to insure they do not short to each other or to any other circuit point. Battery is not installed for alignment.
f. Monitor voltage normally provided to battery with a Digital Voltmeter (DVM) by attaching positive lead of DVM to E11 and negative lead to E19 on power supply pwb.
g. Apply 5 Volts dc (Vdc) to TB2-1 (positive) and TB2-2 (return).
h. Plug AC power plug into a 117 ±5 VAC source, and turn on AA power switch. Allow at least 10 minutes for warm up.
i. Observe voltage on DVM. It shall be 27.6 ±0.3 Vdc. If not, reselect R1 on power supply pwb. Repeat until voltage is 27.6 ±0.3 Vdc.
j. Turn off AA power switch and remove AC plug.
k. Place jumper from terminal E7 to E11 on power supply pwb.
l. Attach probe of oscilloscope to junction of resistor R6 and anode of SCR Q5 on power supply pwb. Attach oscilloscope ground to circuit common such as E19 on power supply pwb or TB2-2.
m. Set a second power supply (PS2) for an output of 24.0 ±0.1 Vdc, turn it off and connect positive lead to E11 and negative to circuit common.
n. Adjust pot R11 fully counterclockwise.
o. Turn on PS2.

NOTE: 5 Vdc power supply should still be on.

p. Slowly adjust R11, observing oscilloscope carefully. Initially, scope shall show a voltage of 24.0 ±0.2 Vdc. As R11 is adjusted, the voltage may decrease slightly, then shall suddenly decrease to less than 1 volt. Do not adjust further.
q. Observe position of pot (R11) and turn off PS2.
r. Repeat steps (n) through (p), being careful to adjust the pot very slowly near the transition point.
s. Reduce voltage of PS2 to less than 20 Vdc, then turn it off.
t. Monitoring DVM and oscilloscope, turn on PS2 and slowly increase voltage from 20 Vdc until the voltage on the scope suddenly

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decreases from the supply voltage to less than 1 Vdc. record DVM reading when this occurs. It shall be 24.0 ±0.2 Vdc. If not, repeat steps (m) through (t) as necessary until the transition occurs at 24.0 ±0.2 Vdc.

u. Turn off PS2 and remove jumpers and instruments added in steps (k) through (m).

v. Plug in AC power plug and turn AA power switch on. Allow sufficient time for warmup (nominally 10 minutes). Record voltage on DVM. If not within tolerance of step (i), reselect R1 as necessary and repeat adjustments of steps (j) through (t). Repeat as necessary until voltage is within tolerance.

w. Affix R1 and seal pot R11 in accordance with Technical Data Package.

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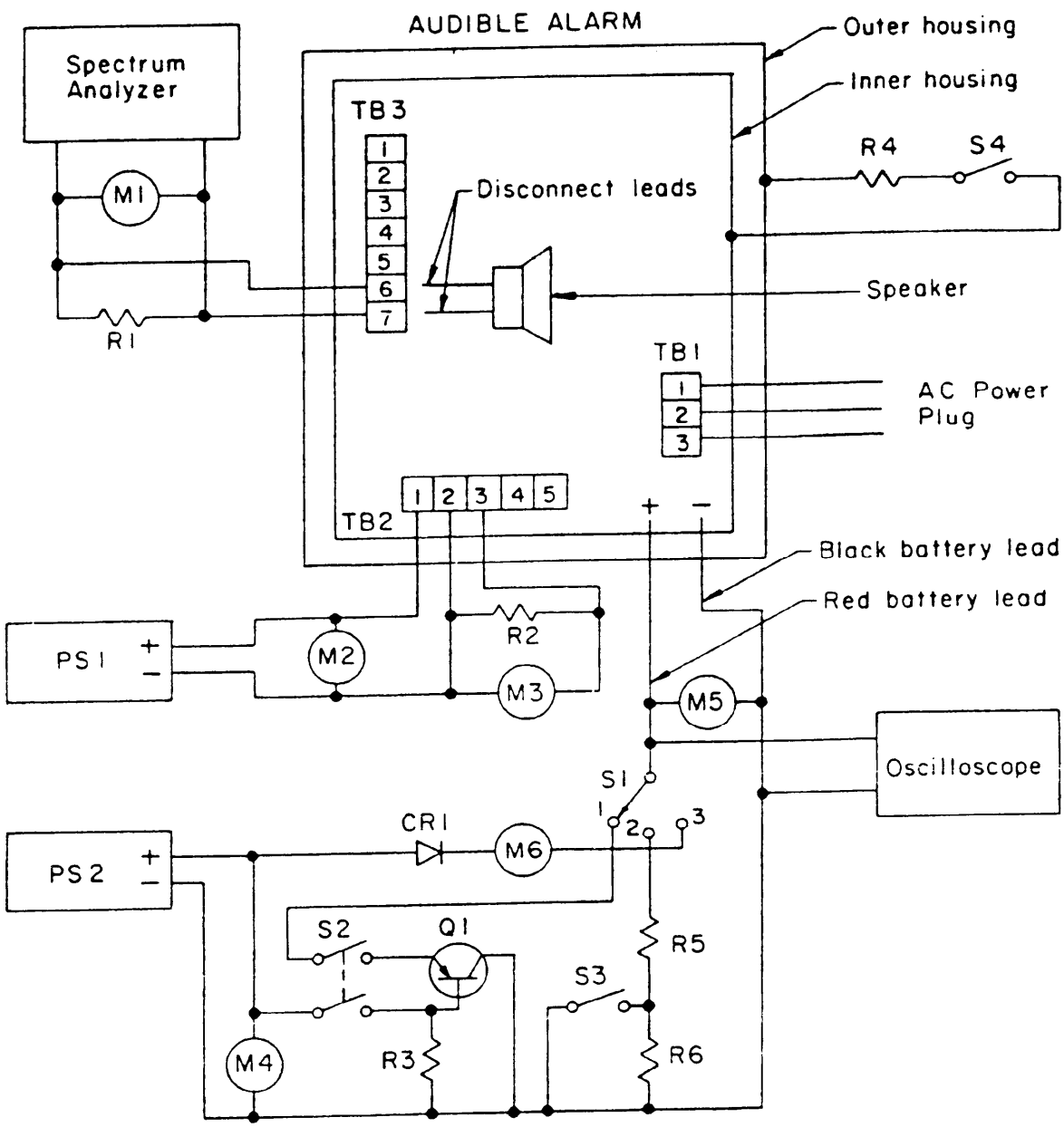


FIGURE 1. Functional test setup (test method 1).

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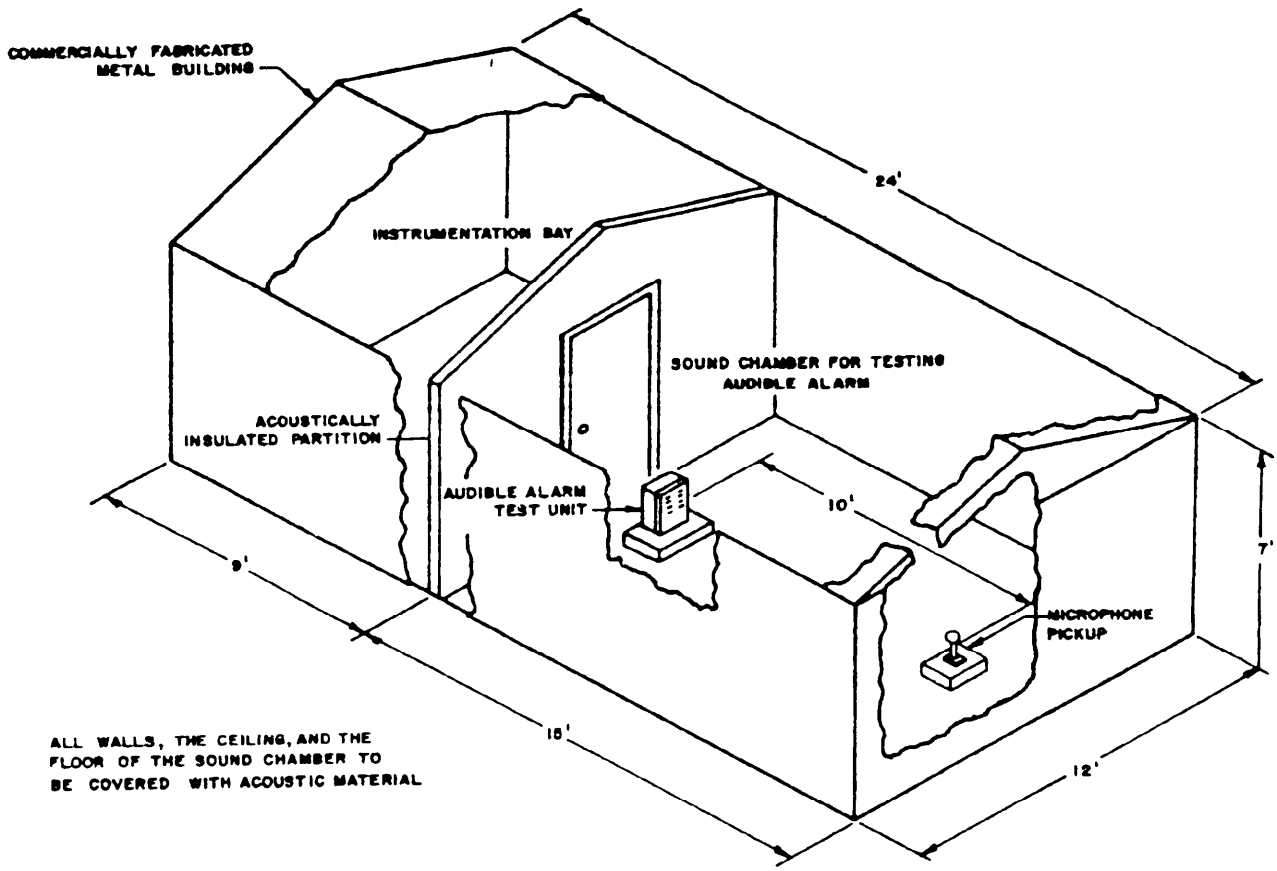


FIGURE 2. Test chamber (test method 2).

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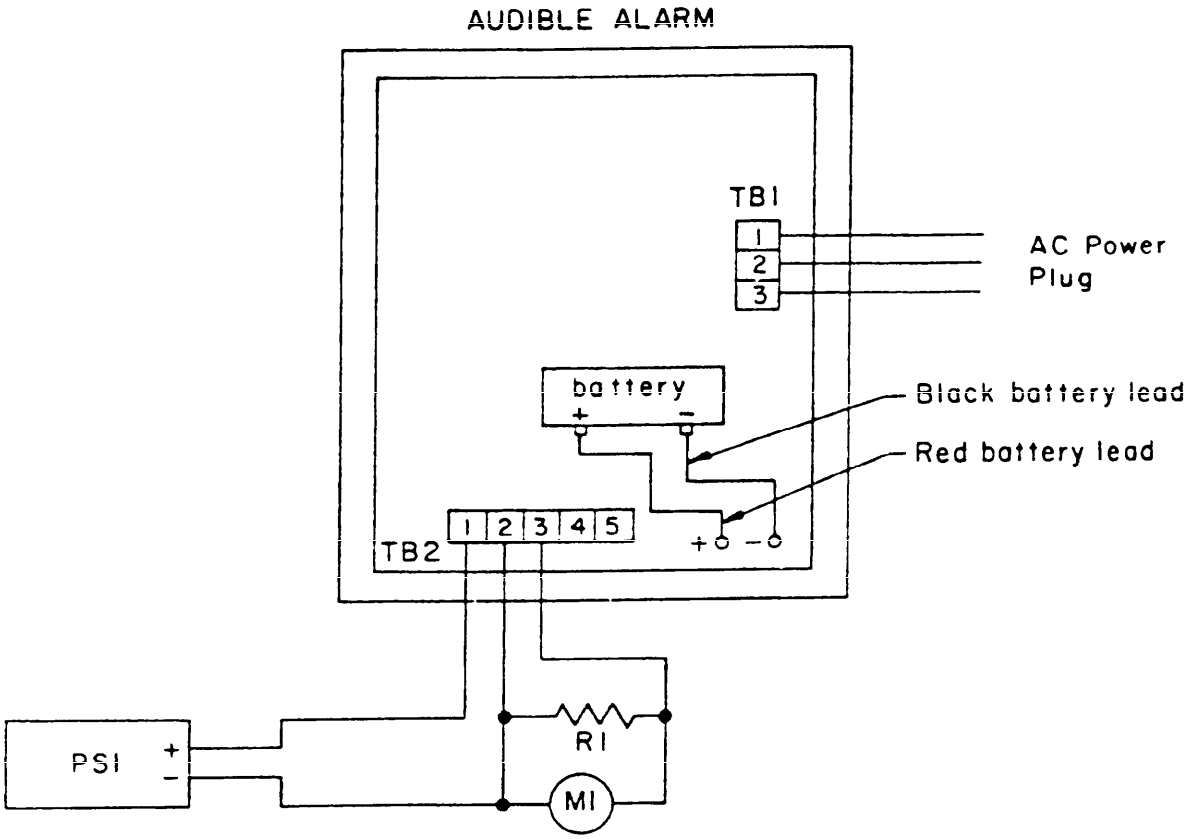


FIGURE 3. Tamper switch tolerance test setup (test method 3).

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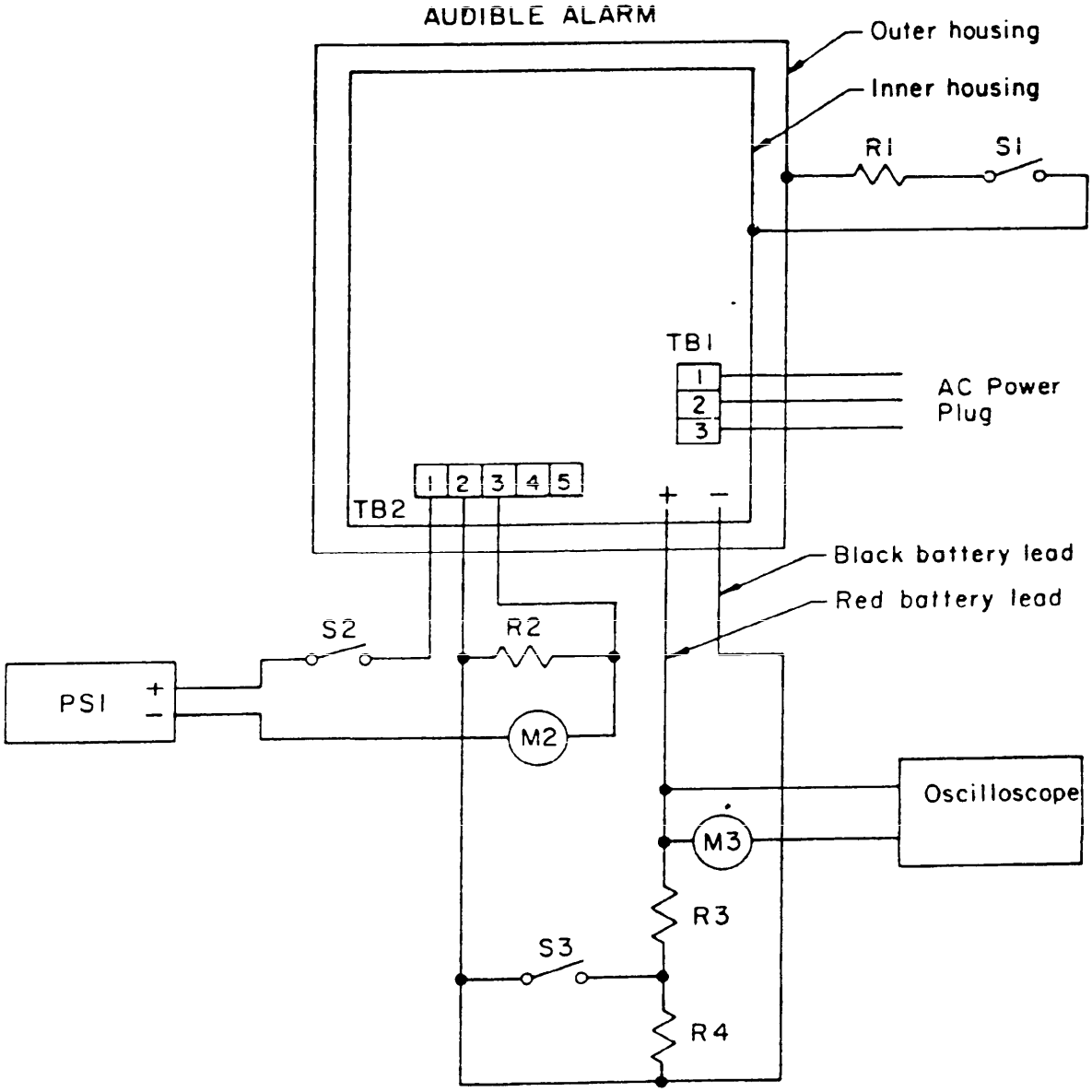


FIGURE 4. Reliability test setup (test method 4).

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Alarm, Audible BZ-204/FSS-9(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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